

from a laser consists of one colour only. As of now, laser light is available at numerous wavelengths in the entire spectrum from ultra-violet (UV) to infrared (IR).

Coherence. Another interesting and important property of laser beam is that the wavefronts of the laser light vibrate in the same phase (see Fig. 3). This is known as coherence and the beam is called a coherent beam. This can be simply explained by the fact that in stimulated emission, a photon of the same frequency induces another excited atom to produce another photon, thus resulting in the emission of two photons that are in phase. Laser beam consists of such photons, which makes it coherent.

There are two types of coherence, namely, temporal and spatial. Temporal coherence measures how long light waves remain in phase as they travel. It describes the correlation or predictable relationship between coherent photons observed at different moments in time. Spatial coherence is used to measure the area over which light is coherent. In other words, spatial coherence describes the correlation between photons at different points in space.

Directionality. It signifies extremely small spread of the laser beam in the direction orthogonal to the direction of travel (see Fig. 4). Technically, the beam divergence of an electromagnetic beam is an angular measure of the increase in beam diameter with distance from the optical aperture. Simply speaking, it is the extent of beam spread perpendicular to the direction of propagation of the beam. Due to the fact that the beam divergence of laser light is very small, it is unidirectional.

Intensity. Light emitted from a laser has very high intensity as compared to any other source of light (see Fig. 5). The unit of intensity is power/unit area/unit solid angle. For example, the intensity of a 1-watt laser light of average quality is 10^7 times higher than the intensity of a 1-watt bulb. This will give you a rough idea as to how intense a laser beam is.



Fig. 5: High-intensity laser beam

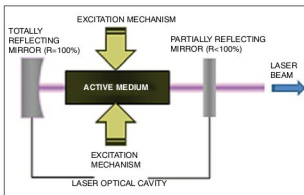


Fig. 6: Components of a laser

Laser components

A laser consists of a gain medium inside a highly reflective optical cavity. The gain medium is a material with properties that allow it to amplify light by stimulated emission. A cavity consists of two mirrors arranged such that light bounces back and forth, each time passing through the gain medium. Typically, one of the two

mirrors, the output coupler, is partially transparent. The output laser beam is emitted through this mirror as shown in Fig. 6.

Modes of laser operation

Lasers can be basically classified as operating in either continuous mode or pulsed mode,



Fig. 7: He-Ne laser



Fig. 8: Ruby laser experiment

which depends on whether their power output is essentially continuous over time or it takes the form of pulses of light on one or another time scale.

Continuous mode of operation. In continuous mode of operation, the output of the laser beam is constant over time. In most cases these lasers